

Javascript

Theory

- **Operators**

- **Unary** → works on one operand (++ , --).
- **Binary** → works on two operands (+, -, *, /, %).
- **Ternary** → shorthand conditional (condition ? trueValue : falseValue).
- **Relational** → compare values (>, <, ==, ===, !=, !==).
- **Logical** → combine conditions (&&, ||, !).
- **Assignment** → update values (+ =, - =, * =, / =, % =).
- **Bitwise** → operate on binary representation (&, |, ^, <<, >>).

- **Control Structures**

- **if / if-else / else-if** → conditional branching.
- **switch** → multiple condition checks.
- **nested if** → condition inside another condition.

- **Loops**

- **for loop** → runs a block for a fixed number of times.
- **while loop** → runs while condition is true.
- **do-while loop** → runs at least once, then checks condition.

Coding Round Questions

- Unary operators → Write a program to demonstrate ++ and -- operators.
- Arithmetic operators → Calculate sum, difference, product, quotient, and remainder of two numbers.
- Relational operators → Compare 10 and "10" using == and ===.
- Logical operators → Write a program to check if a number is between 10 and 20 using &&.
- Assignment operators → Demonstrate += and -= with a variable.
- Bitwise operators → Show the result of 6 ^ 3 and 5 << 1.
- If-else → Write a program to check if a person is eligible to vote.
- Switch case → Print grade based on marks using switch.

- For loop → Print numbers from 1 to 10 using for.
- While loop → Print numbers from 1 to 10 using while.
- Do-while loop → Print numbers from 1 to 10 using do-while.

⚡ Machine Coding Round Challenge

Problem: Student Marks Evaluation System

Requirements:

1. Take input marks for a student.
2. Use **if-else** to assign grades:
 - $\geq 90 \rightarrow A$
 - $\geq 80 \rightarrow B$
 - $\geq 70 \rightarrow C$
 - else $\rightarrow$ Fail
3. Use **switch** to print remarks:
 - A $\rightarrow$ "Excellent"
 - B $\rightarrow$ "Good"
 - C $\rightarrow$ "Average"
 - Fail $\rightarrow$ "Needs Improvement"
4. Use **loops** to print marks of 5 subjects.
5. Use **operators** to calculate total and average marks.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Operators</title>
</head>
<body>

  <h1>Operators</h1>

  <script>
    console.log("Operators");

    // operators =>
    // it's a special symbol that tells the
```

```

// compiler to do specific task

// unary , binary , ternary

// unary => ++, -- (one operand)
// let a = 10;
// a++;
// a--;
// console.log(a--);
// a--
// console.log(a);

// binary => (atleast need two operand)

// arithmetic => (+, -, /, *, %)
// console.log(a+b);
// console.log(a - b);
// console.log(a * b);
// console.log( b / a);
// console.log(7 % 3);

let a = 10;
let b = "10"

// console.log(typeof a);

// relational => (> , <, >=, <=, ==, !=, ===, !==)

// console.log( a > b );
// console.log(a < b );
// console.log(a >= b);
// console.log(a <= b);

// console.log(a == b ); // true checks for value

// console.log(a != b );

// console.log(a === b); //check value and their type

// console.log(a !== b);

// 10 < 30

// 10 > 30 => false

```

```

    // logical ( && || !)
    // && => truth table
    // console.log(true && false);
    // console.log(false || false);

    // console.log(!false);

    // assignment (+, -=, *=, /=, %=)

    // let marks = 90;

    // marks += 5; // 95

    // console.log(marks);

    // bitwise (& | >> << ^)

    console.log(6 ^ 3); // xor
    // same => false
    // different => true

    console.log( 5 << 1 );

</script>
</body>
</html>

```

```

<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Condition</title>
  </head>
  <body>
    <h1>Control structures</h1>

    <script>
      console.log("Control structures");

      // conditional statement

      // if, if-else, else-if, nested if, switch
    </script>
  </body>
</html>

```

```
// let age = 10;

// if( age > 18 ){
//     console.log("you are eligible for vote");
// }else{
//     console.log("you are not eligible to vote");
// }

// else if
// checks multiple condition

// let marks = 30;

// if(marks >= 90){
//     console.log("A Grade");

// }else if(marks >=80){
//     console.log("B Grade");

// }else if (marks >= 70){
//     console.log("C Grade");
// }else{
//     console.log("Fail");
// }

// switch(marks){

//     case 90 :
//         console.log("A Grade");
//         break;

//     case 80 :
//         console.log("B grade");
//         break;

//     default :
//         console.log("Fail");
//         break;
// }

// nested if

// let age = 20;
// let voterId = true;

// if (age > 18) {
//     if (voterId) {
//         console.log("You are eligible for vote");
//     }
// }
```

```
//    }
//  }

// loops for , while, do-while

// for (let i = 1; i <= 10; i++) {
//   console.log(i);
// }

// while loop

let j = 1;

while (j <= 10) {
  console.log(j);
  j++;
}

// do-while loop (
// atleast runs once even the
// condition is false
let x= 1;
do{
  console.log(x);
  x++;
}while(x <= 10);

</script>
</body>
</html>
```